

DBACT THEOREM 1

Proofs and Supporting Results

Boundary-measure-induced Local-CVT practical stability near a selected
nondegenerate local branch

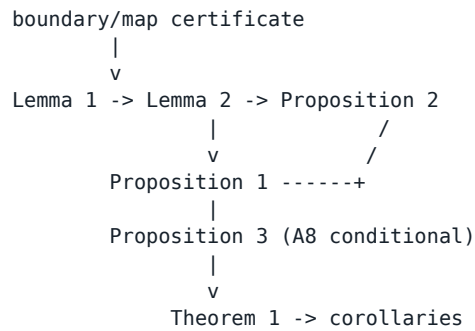
Branch	DBACT-research-v3
Baseline HEAD	64e1905e092d19fd9dcbf0e77fbd39f55f0562ba
Proof status	Conditional local theorem; A8 interval certificate not supplied
Wolfram	Connector exact checks passed; 9 local scripts exit 0
Regression	470 passed, 3 skipped

CLAIM BOUNDARY

No global PL. No formal caging. No finite-time phase completion. No unconditional safety or cooperative-transport success.

Contents and dependency chain

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The chain contains no SEARCH, caging, contact, CBF-invariance, or transport-success node.

1. Scope and frozen claim

Bounded local-map disagreement and bounded reference variation imply practical stability of the boundary-induced Local-CVT allocation dynamics around a nondegenerate local CVT branch.

The system is first order, $p_i' = u_i$. The nominal ENCLOSE law is $u_i^{\text{nom}} = -k_c(p_i - c_{\text{hat}_i})$. The ideal objective uses a saturated performance function; dropping saturation would introduce an uncanceled moving-boundary term.

$$F_R(r) = \min(r^2, R_l^2)$$

$$H^*(P, t) = \int_D \min_i F_R(\|q - p_i\|) \phi^*(q, t) dq$$

Deliberate exclusions

- no global CVT optimality or arbitrary-shape global convergence;
- no finite-time SEARCH-to-HOLD guarantee;
- no formal geometric or topological caging;
- no contact-transport theorem or unconditional cooperative-transport success;
- no safety claim outside feasibility and deviation contracts;
- no uniform Hessian bound for an unspecified local branch.

2. Notation and assumptions

D is a fixed rectangle; P stacks N planar robot positions. $\Gamma(t)$ is the true boundary with perimeter at most P_{max} . The true offset target pushes arc length forward to nu^*_t ; ϕ^* is its convolution with the unnormalized Gaussian plus ϕ_0 .

$$K_{\text{sigma}}(x) = \exp(-\|x\|^2/(2 \text{sigma}^2))$$

$$nu^*_t = \int_{\Gamma} \phi^*(q, t) ds$$

$$\phi^*(q, t) = \phi_0 + (K_{\text{sigma}} * nu^*_t)(q)$$

$$V_i^l = \text{Vor}_i(P) \cap D \cap B(p_i, R_l)$$

$$\|K_{\text{sigma}}\|_1 = 2 \pi \text{sigma}^2$$

$$\|\text{grad } K_{\text{sigma}}\|_1 = \sqrt{2} \pi^{3/2} \text{sigma}$$

$$\|\text{grad } K_{\text{sigma}}\|_{\text{inf}} = \exp(-1/2)/\text{sigma}$$

A1-A10

- A1. D is rectangular and the trajectory stays in fixed compact tube X .
- A2. Separation is at least $d_s > 0$ and $R_{\text{comm}} \geq 2 R_l$.
- A3. The closed piecewise- C^2 boundary is rigid in version 1 and $P_{\Gamma} \leq P_{\text{max}}$.
- A4. $\phi_0 > 0$, $\text{sigma} > 0$, and the stated unnormalized Gaussian is used.
- A5. A conservative map certificate supplies every boundary/normal/offset/voxel/weight/tail bound.
- A6. LocalCVT uses midpoint cells and $\eta_m < m_{\text{min}}$.
- A7. Density biases are disabled or explicitly included in δ .
- A8. A symmetry-reduced selected branch has uniform reduced Hessian $\geq m_H I$ (conditional here).
- A9. Degenerate tie/threshold sets have measure zero; the solution remains on the branch tube.
- A10. ZOH, numerical, tracking, and safety deviations have stacked-norm certificates.

3. Lemma 1 - boundary/map to density

Introduce an exact discrete intermediate measure. The comparison is performed after convolution; no total-variation convergence between a continuous arc-length measure and a Dirac measure is asserted.

```

nu*_t -> nu*_{h,t} = sum_k Delta s_k delta_{xi*_k}
        -> nu_hat_{i,t} = sum_k w_hat_ik delta_{xi_hat_ik}

epsilon_xi = epsilon_b + epsilon_d + 2 d_max sin(epsilon_n/2)
E_w = L_miss + M_spur + sum_matched |w_hat_k - Delta s_k|

epsilon_phi = L_K,1 P_max (epsilon_xi + sqrt(2)v + L_T h_s/2)
              + 2 pi sigma^2 E_w + epsilon_tail          (L1)
epsilon_tail <= pi R_l^2 M_drop exp(-s_sigma^2/2)

```

Proof

Unit normals separated by angle θ differ by $2 \sin(\theta/2)$; the triangle inequality and a square-voxel diagonal give the target error. The fundamental theorem of calculus along a translation segment, followed by Tonelli, gives $\|K(\cdot - a) - K(\cdot - b)\|_1 \leq \|\text{grad } K\|_1 \|a - b\|$.

On each arc, the midpoint curve approximation costs at most $L_{K,1} L_T \Delta s_k^2/2$. Summing uses $\sum \Delta s_k^2 \leq h_s \sum \Delta s_k \leq h_s P_{\max}$. Matched target displacement costs at most $L_{K,1} P_{\max}(\epsilon_{xi} + \sqrt{2}v)$, and total missing/spurious/weight discrepancy costs $\|K\|_1 E_w$.

Every discarded Gaussian source is at least $s_{\sigma} \sigma$ from every query in the disk. Its pointwise contribution is therefore at most $\exp(-s_{\sigma}^2/2)$; summing dropped mass and integrating over at most πR_l^2 proves the tail. The triangle inequality proves (L1).

Evidence boundary

Wolfram exactly checked the kernel integrals and symbolically checked the angle and step identities. Pushforward, translation, curve quadrature, measure decomposition, and tail control are the human analytic proof above.

4. Lemma 2 - mass and centroid perturbation

```

r0 = min(R_l, d_s/2, W_D/2, H_D/2)
m_min = phi_0 pi r0^2 / 4 > 0
m_max = pi R_l^2 (phi_0 + P_max)
phi_hat_max = phi_0 + P_max + E_w
L_hat_phi <= (P_max + E_w) exp(-1/2)/sigma

r_h = h/sqrt(2), A_V <= pi R_l^2, P_V <= 2 pi R_l
eta_m = L_hat_phi r_h A_V + phi_hat_max(2 P_V r_h + pi r_h^2)
eta_a = (phi_hat_max+R_l L_hat_phi)r_h A_V
        + R_l phi_hat_max(2 P_V r_h + pi r_h^2)
epsilon_q = (eta_a + R_l eta_m)/(m_min - eta_m)

||c_hat_i-c*_i|| <= 2 R_l epsilon_phi_i/m_min
                  + epsilon_q_i + epsilon_G_i      (L2)

```

Proof

Separation puts $B(p_i, d_s/2)$ in robot i 's Voronoi cell. The intersection with the local disk and rectangle contains at least a quarter disk of radius r_0 ; $\phi_i \geq \phi_0$ gives $m_{\min}$. Kernel boundedness and disk area give the ideal $m_{\max}$. Lemma 1 bounds mapped total weight by $P_{\max} + E_w$; differentiating its convolution gives the displayed mapped-density bounds.

A midpoint cell entirely inside the exact convex cell contributes at most $L_{\hat{\phi}} r_h$ times area. Boundary-cut cells occupy a parallel strip of area at most $2P_V r_h + \pi r_h^2$. This proves η_m . Repeating the argument for the shifted first moment $a = \int (q - p_i) \phi dq$ proves η_a .

Because $m_{\hat{\phi}} \geq m_{\min} - \eta_m > 0$, the ratio identity gives ϵ_q . Lemma 1 changes continuum mass by at most ϵ_{ϕ} and shifted moment by at most $R_l \epsilon_{\phi}$; both densities retain the base-density floor, giving $2R_l \epsilon_{\phi}/m_{\min}$. Add the separately certified geometric-cell term. The denominator gate fails closed.

5. Proposition 1 - ideal gradient and descent

$$\begin{aligned} \text{grad}_{\{p_i\}} H^* &= 2 m_i^* (p_i - c_i) & (P1-a) \\ u_i^* &= -k_c(p_i - c_i) = -[k_c/(2m_i^*)] \text{grad}_{\{p_i\}} H^* \end{aligned}$$

Inside V_i^l the active integrand derivative is $2(p_i - q)$; outside the disk it is zero because the cost is saturated. On a shared Voronoi face, competing minimum values agree and the two fluxes cancel. On $r=R_l$, the inner and outer costs both equal R_l^2 , so the circle flux cancels. The workspace boundary is fixed and the exceptional tie/threshold sets have measure zero. Only the volume derivative remains, proving P1-a.

$$\begin{aligned} A(P, t) &= \text{diag}_i(k_c/(2m_i^*) \ I_2) \succeq a_0 \ I \\ a_0 &= k_c/(2m_{\max}) \\ \text{Hdot}^* &= -\text{grad } H^{*T} A \text{ grad } H^* \preceq -a_0 \ ||\text{grad } H^*||^2 \end{aligned}$$

If a point in $B(p_i, R_l)$ is won by robot j , then $||p_j - p_i|| \leq 2R_l \leq R_{\text{comm}}$. Thus the communication set contains every local competitor and the cell is exact. The Wolfram fixed-square gradient check is only a sanity example; it is not the shape-derivative proof.

6. Proposition 2 - aggregate disturbance

$$\begin{aligned} \text{Pdot} &= -A(P, t) \text{ grad } H^*(P, t) + d(t) \\ ||d(t)|| &\leq \delta \\ \delta &= k_c \sqrt{\sum_i \epsilon_{c_i}^2} + \epsilon_{\text{zoh}} + \epsilon_{\text{num}} \\ &\quad + \epsilon_{\text{trk}} + \epsilon_{\text{sf}} & (P2) \end{aligned}$$

Lemma 2 bounds the centroid component, and the stacked triangle inequality adds the remaining deviations. Production commands use a per-agent Euclidean speed cap, so $||u_i^{\text{sf}} - u_i^{\text{nom}}|| \leq 2u_{\max}$ and $\epsilon_{\text{sf}} \leq 2\sqrt{N}u_{\max}$. A componentwise box cap would require $2\sqrt{2N}u_{\max}$. The certificate records the chosen convention.

7. Proposition 3 - reference rate and regularity

$$\begin{aligned}
 x_i &= b + d \, n, & x_{i,\text{dot}} &= b_{\text{dot}} + d_{\text{dot}} \, n + d \, n_{\text{dot}} \\
 v_{x_i,\text{max}} &= v_{0,\text{bar}} + \omega_{0,\text{bar}}(R_0 + d_{\text{max}} + L_d) \\
 \nu_{\phi} &\leq L_K, l \, P_{\text{max}} \, v_{x_i,\text{max}} \\
 \nu_H &= 2 \, R_l^2 \, \nu_{\phi}
 \end{aligned} \tag{P3}$$

The rigid-twist bound controls every target speed; convolution translation controls the density rate. Since $0 \leq \min_i F_R \leq R_l^2$, $|\partial_t H^*| \leq R_l^2 \, \nu_{\phi}$. On fixed compact X , $|\min f - \min g| \leq \sup |f - g|$ supplies the same bound for $H_{\min}^*$, producing ν_H . Deformation, perimeter change, or split/merge events require a new measure-rate assumption.

M5 decision

$$\begin{aligned}
 \frac{1}{2} \|\text{grad } H^*\|^2 &\geq \mu \, (H^* - H_{\min}^*) \\
 H^* - H_{\min}^* &\geq (\alpha/2) \, \text{dist}^2(P, C^*(t))
 \end{aligned}$$

A uniform reduced Hessian lower bound on a convex symmetry-reduced chart implies both relations with $\mu = \alpha = m_H$ (or smaller constants). The implication is proved. Its premise is conditional: no selected branch, gauge, cell combinatorics, or complete parameter tube was supplied for validated interval subdivision. No floating-point eigenvalue is used as proof.

8. Theorem 1 - local practical stability

$$\begin{aligned}
 V(P,t) &= H^*(P,t) - H_{\min}^*(t) \geq 0 \\
 a_0 &= k_c / (2m_{\max}), \quad \lambda_H = a_0 \mu \\
 B &= \delta^2 / (2a_0) + \nu_H \\
 V(t) &\leq \exp[-\lambda_H(t-t_0)] V(t_0) \\
 &\quad + (B/\lambda_H)(1 - \exp[-\lambda_H(t-t_0)]) \quad (T1-a) \\
 \limsup \text{dist}(P(t), C^*(t)) &\leq \sqrt{2B/(\alpha \lambda_H)} \quad (T1-b)
 \end{aligned}$$

Proof - every inequality named

$$\begin{aligned}
 \dot{V} &= \text{grad } H^*{}^T \dot{P} + \text{partial}_t H^* - H_{\min}^*{}_{\dot{}} \\
 &\leq -\text{grad } H^*{}^T A \text{grad } H^* + \|\text{grad } H^*\| \|\dot{d}\| + \nu_H \\
 &\quad [\text{Propositions 1 and 3}] \\
 &\leq -a_0 \|\text{grad } H^*\|^2 + \delta \|\text{grad } H^*\| + \nu_H \\
 &\quad [\text{Proposition 2 and } A \geq a_0 I] \\
 &\leq -(a_0/2) \|\text{grad } H^*\|^2 + \delta^2 / (2a_0) + \nu_H \\
 &\quad [\text{Young}] \\
 &\leq -a_0 \mu V + B = -\lambda_H V + B \\
 &\quad [\text{local PL}]
 \end{aligned}$$

The scalar comparison lemma yields T1-a. Hence $\limsup V \leq B/\lambda_H$. Local quadratic growth gives $\text{dist}(P, C^*) \leq \sqrt{2V/\alpha}$, which yields T1-b. The infinite-time limit is asserted only if the trajectory continues to satisfy the regular-tube premises.

Status

The analytic implication is complete, and Wolfram verifies the scalar algebra. The result remains conditional on A5, A8-A10 and the run/tube certificates. It is local, not global.

9. Corollaries

Frozen perfect information

$\delta=0, \nu_H=0 \Rightarrow B=0$
 $V(t) \leq \exp[-\lambda_H(t-t_0)] V(t_0), \quad \text{dist}(P, C^*) \rightarrow 0$

The generator and unit tests require exactly zero forcing and zero ultimate radius in this case.

Sampled data

$P_{k+1} = P_k + dt[-A_k g_k + d_k], \quad a_0 \leq A_k \leq a_1$
 $\bar{a} = a_0 - (L_H a_1^2 dt)/2$
 $C_\delta = \delta(1 + L_H a_1 dt)$
 $B_d = C_\delta^2/(2\bar{a}) + (L_H/2)\delta^2 dt + \nu_H$
 $dt < 2a_0/(L_H a_1^2), \quad 0 < \bar{a} \mu dt \leq 1$
 $V_{k+1} \leq (1 - \bar{a} \mu dt)V_k + B_d dt$
 $\limsup \text{dist}(P_k, C_k^*) \leq \sqrt{2B_d/(\alpha \bar{a} \mu)}$

The descent lemma first gives $dt[-a_0 x^2 + \delta x] + (L_H dt^2/2)(a_1 x + \delta)^2 + \nu_H dt$. Expansion, Young, and local PL give the recursion. The geometric series and quadratic growth give the ultimate radius. Wolfram checked the expansion; randomized positive-parameter tests independently compare the direct one-step expression with the certified upper bound.

10. Limits and evidence status

Result	Status	Limiting condition
Lemma 1	PROVED + CAS	A5 map bounds and tail separation
Lemma 2	PROVED + CAS	rectangle, midpoint gate, exact-cell contract
Proposition 1	PROVED	regular tie/threshold sets; saturated cost
Proposition 2	PROVED	declared stacked disturbance components
Proposition 3	CONDITIONAL	rate proved; A8 branch coercivity assumed
Theorem 1	CONDITIONAL	complete implication under A1-A10
Sampled corollary	PROVED	explicit step gates

The illustrative certificate validates the dependency wiring and gates. Its numeric radius is intentionally reported as conservative and may be vacuous; it is not experimental evidence and is not used to claim transport performance.

Evidence classes and runtime gates

- PROVED: the analytic implication is written in this supplement.
- VERIFIED: Wolfram reproduced only the identified integral or algebraic subclaim.
- CONDITIONAL: a named branch, tube, map bound, or runtime deviation certificate remains required.
- NUMERIC-SANITY: debugging evidence only; no such result is used as proof here.
- Fail-closed gates: locality, positive arc length, midpoint rule, $\eta_m < m_{\min}$, disabled unbudgeted biases, and sampled step size.

Claim audit

The supplement deliberately contains no global PL statement, no formal caging claim, no finite-time phase-completion result, and no unconditional transport or safety guarantee.

Appendix A. Constant ledger

Constant	Value	Unit	Status
kernel_l1	0.251327	m^2	verified
kernel_grad_l1	1.57496	m	verified
kernel_grad_linf	3.03265	1/m	verified
epsilon_xi	0.00449999	m	conditional
epsilon_phi	0.106034	m^3	proved
m_min	5.02655	m^3	proved
m_max	34.5827	m^3	proved
grid_spacing	0.000390625	m	verified
eta_m	0.0598959	m^3	proved
eta_a	0.0574695	m^4	proved
epsilon_q	0.0212188	m	proved
epsilon_c	0.0549703	m	proved
delta	0.156932	m/s	conditional
nu_phi	0.464928	m^3/s	proved
nu_H	0.595108	m^5/s	proved
a0	0.0130123	$1/(s*m^3)$	proved
lambda_H	0.000650615	1/s	conditional
B	1.54143	m^5/s	proved
ultimate_radius	307.843	m	conditional
sampled_Bd	1.54617	m^5/s	proved
sampled_ultimate_radius	308.792	m	conditional

```
(epsilon_b, epsilon_n, epsilon_d, v, h_s, E_w, M_drop)
-> epsilon_phi -> epsilon_c -> delta -> B
-> sqrt(2B/(alpha lambda_H))
```

Appendix B. Wolfram verification manifest

Connector: Wolfram (version not exposed). Local runtime: WolframScript 1.14.0. All nine included scripts executed with exit code 0. Exact connector outputs and assumptions are archived in `theory/wolfram/results`.

ID	Target	Evidence	Verdict
W-L1-01/02	Gaussian L1 integrals	EXACT	PASS
W-L1-03	gradient radial maximum	SYMBOLIC	PASS
W-L1-04/05	normal/step identities	SYMBOLIC	PASS
W-L2-01/02	ratio, monotonicity, mass	SYMBOLIC	PASS
W-P1-01	integrand and fixed-cell check	SYMBOLIC	PASS
W-P3-01	reference-rate algebra	SYMBOLIC	PASS
W-T1-01	Young and comparison ODE	SYMBOLIC	PASS
W-C1-01	sampled-data expansion	SYMBOLIC	PASS
W-M5-01	generic quadratic implication	SYMBOLIC	PASS
W-M5-02	actual branch Hessian box	INTERVAL	CONDITIONAL

Evidence rule: exact/symbolic output supports only the stated integral or algebra. Interval evidence would need to cover the complete selected tube. Numeric sanity is debugging evidence only. There are no numeric-sanity entries in this package.

Reference

J. Cortes, S. Martinez, and F. Bullo, 'Spatially-distributed coverage optimization and control with limited-range interactions,' ESAIM: COCV, 11(4):691-719, 2005. DOI 10.1051/cocv:2005024.